College of Engineering, Construction and Living Sciences
Bachelor of Information Technology
ID608001: Intermediate Application Development Concepts
Level 6, Credits 15

Project

Assessment Overview

In this **individual** assessment, you will design and develop **two** games.

Learning Outcomes

At the successful completion of this course, learners will be able to:

1. Apply design patterns and programming principles using software development best practices.
2. Design and implement full-stack applications using industry relevant programming languages.

Assessments

Assessment	Weighting	Due Date	Learning Outcomes
Practical	20%	Sunday 19th October at 11.59 PM	1
Project	80%	Sunday 9th November at 11.59 PM	1 and 2

Conditions of Assessment

You will complete this assessment mostly during your learner-managed time. However, there will be time during class to discuss the requirements and your progress on this assessment. This assessment will need to be completed by **Sunday 9th November** at **11.59 PM**.

Pass Criteria

This assessment is criterion-referenced (CRA) with a cumulative pass mark of **50%** over all assessments in **ID608001: Intermediate Application Development Concepts**.

Authenticity

All parts of your submitted assessment **must** be completely your work. Do your best to complete this assessment without using AI tools. You need to demonstrate to the course lecturer that you can meet the learning outcomes for this assessment.

Learning to use AI tools is an important skill. While AI tools are powerful, you **must** be aware of the following:

- If you provide an AI tool with a prompt that is not refined enough, it may generate a not-so-useful response
- Do not trust the AI tool's responses blindly. You **must** still use your judgement and may need to do additional research to determine if the response is correct
- Acknowledge what AI tool you have used. In the assessment's repository **README.md** file, please include what prompt(s) you provided to the AI tool and how you used the response(s) to help you with your work

It also applies to code snippets retrieved from **StackOverflow** and **GitHub**.

Failure to do this may result in a mark of **zero** for this assessment.

Policy on Submissions, Extensions, Resubmissions and Resits

The school's process concerning submissions, extensions, resubmissions and resits complies with **Otago Polytechnic** policies. Learners can view policies on the **Otago Polytechnic** website located at https://www.op.ac.nz/about-us/governance-and-management/policies.

Submission

You **must** submit all application files via **GitHub**. Create a repository and add the course lecturer as a collaborator. The latest application files will be used to mark against the marking rubric. Please test before you submit. Partial marks **will not** be given for incomplete functionality. Late submissions will incur a **10% penalty per day**, rolling over at **12.00 AM**.

Extensions

Familiarise yourself with the assessment due date. Extensions will **only** be granted if you are unable to complete the assessment by the due date because of **unforeseen circumstances outside your control**. The length of the extension granted will depend on the circumstances and **must** be negotiated with the course lecturer before the assessment due date. A medical certificate or support letter may be needed. Extensions will not be granted on the due date and for poor time management or pressure of other assessments.

Resits

Resits and reassessments **are not** applicable in **ID608001: Intermediate Application Development Concepts**.

Functionality - Learning Outcomes 1, 2 (25%)

- **Godot 2D Platformer** game with the following functionality:
  - Player movement using keyboard input.
  - Platform collision system, i.e., solid, moving and one-way platforms.
  - Health system.
  - Win and lose conditions.
  - Three distinct levels with increasing difficulty.
  - Hazards and obstacles, i.e., spikes, traps or pits.
  - Three distinct enemies with different behaviours.
  - Double jump ability.
  - Collectible items, i.e., coins or power-ups.
  - Scoring system.
  - Choose five features of your choice. These features should be different from the ones listed above.
- **3D Unity/Godot Obstacle Dodging** game with the following functionality:
  - Player movement using keyboard or mouse input.
  - Obstacle collision system, i.e., solid and moving obstacles.
  - Health system.
  - Five distinct obstacles with different behaviours.
  - Two distinct levels with increasing difficulty.
  - Collectible items, i.e., coins or power-ups.
  - Scoring system.
  - Camera effects, i.e., screen shake.
  - Choose five features of your choice. These features should be different from the ones listed above.
- Deploy both games to **Itch.io**.

Code Quality and Best Practices - Learning Outcomes 1, 2 (40%)

- Write clean, readable, and maintainable code.
- Use modular, reusable components and avoid code duplication.
- Keep code simple and easy to understand.
- Optimise performance by managing resources efficiently.

Version Control - Learning Outcome 1 (5%)

- Regular commits are made throughout development, demonstrating meaningful progress.
- Commit messages are clear and descriptive.
- The repository shows a clean and logical commit history that reflects the development process of the games.

Reflection - Learning Outcomes 1, 2 (5%)

- Write a short reflection (300-500 words) on your development process and learning throughout the project, including:
  - Challenges you faced and how you overcame them.
  - What you would do differently in future projects.
  - New skills, tools or practices you learned during the project.

Presentation - Learning Outcomes 1, 2 (5%)

- Record a 5-10 minute screen recording of your games.
- Demonstrate the **Godot 2D Platformer** game and **3D Unity/Godot Obstacle Dodging** game features.
- Make sure the video is clear and easy to follow.
- Submit a link to the presentation in your repository's **README.md** file.